

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037
<https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Software Engineer with **18 years** of end-to-end ownership of large-scale distributed systems, including the YouTube Live streaming pipeline, Google Ads real-time bidding platform, and Microsoft Azure DevOps CI/CD service. Proven expertise in **C++**, **Java**, **Python**, **Kubernetes** and **AWS**. Delivered a 45% reduction in deployment time while supporting a 3× increase in request volume. Regularly on-call, collaborating with product, security and SRE partners. Next focus: building a self-service, kernel-level platform for cloud resource provisioning. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Programming Languages: C++, Java, Python, Go, Shell, TypeScript, JavaScript, HTML, CSS, SQL

Cloud Platforms: AWS, GCP, Azure, Google Cloud Pub/Sub, Amazon S3

Container & Orchestration: Kubernetes, Docker, Helm, Istio, Prometheus

Infrastructure as Code: Terraform, CloudFormation, Ansible, Chef, GitOps

Observability & Reliability: Grafana, Jaeger, OpenTelemetry, SRE Practices, Chaos Engineering

Work Experience

Staff Software Engineer, YouTube (Remote) | 02/2025 - Present

- Led end-to-end redesign of YouTube Live streaming pipeline, migrating core transcoding services to **Kubernetes** on **AWS**, cutting average start-up latency by 30% while handling 2× traffic growth without outage.
- Partnered with product, security and SRE teams to define a self-service platform for cloud resource provisioning, targeting a 2026 launch that will serve 10 k internal teams.
- Built an automated CI/CD workflow with **Terraform** and **GitHub Actions**, decreasing deployment cycle time from 45 minutes to 25 minutes across 150 microservices.
- Implemented a kernel-level resource-allocation module in **C++**, reducing CPU contention by 22% and enabling finer-grained scaling for high-definition streams.
- Introduced a real-time observability stack using **Prometheus** and **Grafana**, improving incident detection speed by 40% and supporting on-call rotation.

Staff Software Engineer, Whatnot (Remote) | 05/2024 - 02/2025

- Refactored legacy payment gateway in **Java** and **Spring Boot**, reducing transaction latency by 18% and cutting operational costs by \$1.2 M annually.
- Architected a multi-region order-processing system on **GCP**, achieving 99.99% availability and supporting a 150% quarterly sales surge.
- Deployed a container-native CI pipeline using **Docker** and **Helm**, cutting build times by 35% for 80+ services.
- Mentored a cross-functional squad of 12 engineers, establishing code-review standards that lowered post-release defects by 27%.
- Integrated **OpenTelemetry** tracing across microservices, providing end-to-end visibility that accelerated root-cause analysis by 50%.

Staff Software Engineer/Manager, Google (Remote) | 03/2015 - 04/2024

- Directed the migration of Google Ads bidding engine from on-prem to **Google Cloud Platform**, increasing bid throughput by 40% while maintaining sub-millisecond latency.
- Designed a high-performance **C++** cache layer that reduced query latency by 28% for 200 M daily requests.
- Established a company-wide **Terraform** module library, standardizing infrastructure provisioning and saving 12 k engineer-hours over 3 years.
- Implemented automated chaos testing with **Chaos Monkey**, uncovering hidden failure modes and strengthening system resilience.
- Co-led an on-call rotation handling 24/7 incident response, achieving a mean-time-to-recovery (MTTR) improvement of 33%.

Software Engineer, Google (Remote) | 02/2011 - 02/2015

- Built a distributed logging pipeline using **Java** and **Kafka**, processing 5 TB of data daily and reducing log ingestion latency by 20%.
- Created a self-service dashboard with **Angular** and **REST APIs**, enabling non-engineers to request compute resources, decreasing request turnaround by 45%.
- Optimized a search indexing service in **Go**, cutting index update time from 12 seconds to 5 seconds.
- Authored internal tooling for automated code quality checks, raising code coverage from 62% to 85%.
- Collaborated with security teams to integrate **OAuth2** across internal APIs, enhancing authentication compliance.

Software Design Engineer II, Microsoft (Remote) | 10/2007 - 12/2011

- Led the redesign of Azure DevOps build agents in **C#**, improving build throughput by 30% and supporting a 2× increase in concurrent jobs.
- Implemented a fault-tolerant deployment orchestrator using **PowerShell** and **Azure Resource Manager**, reducing deployment failures by 22%.
- Developed a monitoring plug-in for **System Center Operations Manager**, providing real-time health metrics for 500+ services.
- Co-authored internal guidelines for secure configuration management, decreasing configuration drift incidents by 15%.
- Participated in on-call rotation, resolving high-severity incidents within SLA 98% of the time.

Software Design Engineer Intern, Microsoft (Remote) | 06/2006 - 08/2006

- Contributed to the prototype of a cloud-based file sync service in **C#**, achieving sub-second sync latency for 10 k test files.
- Presented findings to senior engineers, influencing the adoption of a new caching strategy that later reduced read latency by 18%.
- Automated unit-test generation using **PowerShell**, increasing test coverage by 12% for the assigned module.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007